

NETWORKS AND COMPLEXITY

Exercise Sheet 16: The Critical Brain

*This is an exercise sheet from the forthcoming book *Networks and Complexity*.
Find more exercises and solutions at <https://github.com/NC-Book/NCB>*

Ex 16.1: Geometric dynamics [Lvl. 1]

Starting from the initial condition $\delta(0) = \delta_0$, integrate

$$\dot{\delta} = a\delta^2.$$

Ex 16.2: Fold bifurcation normal form [Lvl. 2]

Compute the steady states and their stability for the system

$$\dot{x} = ax^2 + bp.$$

Ex 16.3: Abstract sensitivity example [Lvl. 2]

You study an ODE that is quite complicated,

$$\dot{x} = -x^4 + 2px^3 + 2p^2x - px^2 + x - 2p$$

However, you know that in the real world system $p = 1$, $x = 2$.

- Verify that $x^* = 2$ is indeed a steady state at $p = 1$.
- Use the implicit function theorem to estimate how the steady state will respond if p is changed a little bit.

Ex 16.4: Degeneracy of the epidemic threshold [Lvl. 2]

Consider the SIS model in the form

$$\dot{x} = f(x, p) = px(1 - x) - x$$

where we have used x instead of I as the variable. Find the epidemic threshold, and then check the genericity conditions of the fold bifurcation $f_p \neq 0$, $f_{xx} \neq 0$.

Ex 16.5: Cellular switch [Lvl. 3]

In human cells, a certain protein can be present in an activated and an inactivated form. The dynamics of the amount of activated protein, x , can be modeled as

$$\dot{x} = \frac{x^2}{1 + x^2}(S - x) - x,$$

where S is the total amount of protein in either form. Biologists suspect that the protein acts as a switch that the cell uses to store information. Is this possible?

Ex 16.6: Transcritical bifurcation normal form [Lvl. 3]

Consider a dynamical system of the form $\dot{x} = f(x, p)$, that has a steady state x^* . At a parameter value p^* the steady state undergoes a bifurcation such that $f_x(x^*, p^*) = 0$. Moreover $f_p(x^*, p) = 0$ for all p .

- a) Find the normal form: Define $\delta = x - x^*$ and $\rho = p - p^*$ then Taylor expand the $\dot{\delta}$ until you have enough terms such that you can argue that all further Taylor terms are negligible if δ and ρ are sufficiently small. (Hint: note that the conditions also imply $f_{pp}(x^*, p) = 0$)
- b) To simplify the notation, define a new variable x , and new parameters a and b to show that your normal form can be written as

$$\dot{x} = ax + bx^2.$$

- c) Compute the steady states and determine their stability. Define a new parameter $p = -a/b$ and draw the bifurcation diagram of x^* over p for the case $b < 0$.

Ex 16.7: Pitchfork normal form [Lvl. 3]

Consider a system $\dot{x} = f(x, p)$ with mirror anti-symmetry such that $f(x, p) = -f(-x, p)$. Show that there is a steady state at $x^* = 0$, then consider a bifurcation of this steady state that occurs at p^* . Follow the steps from Ex. 16.6 to derive the normal form of this bifurcation and analyze it.

Ex 16.8: Spruce Budworm [Lvl. 4]

The spruce budworm is a moth that feeds on the needles of spruce trees, and can occur in large outbreaks. The dynamics is modeled by

$$\dot{x} = x(1 - x/K) - x^2/(1 + x^2)$$

where K is a parameter that indicates the mass of spruce needles in the area. Analyze this system to determine whether this model can explain the sudden outbreaks.

Ex 16.9: Supersensitivity in degenerate bifurcations [Lvl. 4]

In the pitchfork and transcritical bifurcations the slope of the steady state in the bifurcation point is undefined, because there are several branches with different slopes. To find the slope of the nontrivial branch we first have to remove the trivial steady state, which we can do by dividing the equation of motion by x .

- a) Define $g(x, p) = f(x, p)/x$ and then use the implicit function theorem for g to find the slope in the critical point of the pitchfork and transcritical bifurcations. (Hint: you might need to use l'Hospital's rule)
- b) Test your result on the transcritical bifurcation in the SIS-system from Ex. 16.4.

Ex 16.10: Subcritical transcritical bifurcation [Lvl. 4]

We don't usually think of the transcritical bifurcation as having a subcritical and supercritical form. However, in Chap. 15 the transcritical bifurcation behaved in a subcritical way, causing a discontinuous phase transition. Consider the transcritical normal $\dot{x} = ax + bx^2$. Assume that the model demands $x \geq 0$ and find the conditions under which the bifurcation behaves subcritically.

Ex 16.11: An unusual case [Lvl. 4]

Determine the type of the bifurcation that occurs in the system

$$\dot{x} = (x - 1 - p)(2p - x)$$

Check your answer carefully.

Ex 16.12: The complicated example revisited [Lvl. 4]

Let's return to the system from Ex. 16.3. Could you use the approach from the implicit function theorem also to compute $x''(p)$ at $p = 1$? How about $x'''(p)$ and $x''''(p)$? Use this to construct $x^*(p)$ as a Taylor expansion. Use your result to find $x^*(0)$, and verify that it is a steady state.